

BLUESTOCK Mutual Fund Analytics Capstone Project

Final Report

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Table of Contents

Table of Contents.....	2
1. Executive Summary	4
Key Highlights	4
2. Data Sources	5
3. Project Methodology	6
3.1 Tools and Technologies.....	6
4. ETL Design and Data Architecture	7
3.1 Extract	7
3.2 Transform	7
3.3 Load	7
5. Exploratory Data Analysis Findings	8
4.1 NAV Trends and Market Cycles.....	8
4.2 AUM Growth and Fund House Dominance	8
4.3 SIP Inflow Growth.....	8
4.4 Category-wise Inflows.....	8
4.5 Investor Demographics.....	8
4.6 Geographic Distribution	8
4.7 Folio Growth	8
4.8 Return Correlation and Sector Concentration	9
6. Fund Performance Analysis	10
5.1 Returns and CAGR.....	10
5.2 Risk-Adjusted Metrics	10
5.3 Alpha and Beta	10
5.4 Maximum Drawdown	10
5.5 Fund Scorecard	10
5.6 Benchmark Comparison	11
7. Interactive Dashboard	12
7.1 Page 1 - Industry Overview	12
7.2 Page 2 - Fund Performance.....	12
7.3 Page 3 - Investor Analytics	13
7.4 Page 4 - SIP and Market Trends	14
8. Advanced Analytics and Risk Metrics.....	15
8.1 Value at Risk (VaR) and Conditional VaR (CVaR)	15
8.2 Rolling 90-Day Sharpe Ratio	15
8.3 Investor Cohort Analysis.....	15

8.4 SIP Continuity Analysis.....	15
8.5 Fund Recommender	15
8.6 Sector Concentration (HHI).....	15
9. Limitations.....	16
10. Recommendations	16
11. Conclusion	17
Appendix A: Glossary of Key Metrics	18
Appendix B: Project Repository Structure	18
Appendix C: Self-Review Checklist	19

1. Executive Summary

This report presents a complete data analysis of the Indian mutual fund industry, covering fund performance, investor behaviour, risk metrics, and market trends across 40 mutual fund schemes from 10 major fund houses. The analysis spans data from January 2022 to December 2025 and combines daily NAV history, investor transaction records, scheme performance data, and benchmark index movements.

The project was carried out in seven structured phases: data ingestion, data cleaning and database design, exploratory data analysis, fund performance analytics, dashboard development, advanced risk analytics, and final reporting. Each phase built on the previous one, resulting in a complete end-to-end analytics pipeline.

Key Highlights

- Industry-wide AUM reached approximately ₹391.76 lakh crore across the fund houses studied, with SBI Mutual Fund holding the largest share.
- Monthly SIP inflows grew from ₹11,517 crore in January 2022 to an all-time high of ₹31,002 crore in December 2025, a growth of nearly 169% over four years.
- Total investor folio count nearly doubled, growing from 13.26 crore in January 2022 to 26.12 crore by December 2025.
- Small Cap equity funds delivered the strongest 1-year returns among the 40 schemes analysed, with ABSL Small Cap Fund and SBI Small Cap Fund leading the category.
- Liquid funds and short-duration debt funds showed the most stable, low-risk return profiles, reflected in very high Sharpe ratios relative to their volatility.
- Punjab, Tamil Nadu, and Madhya Pradesh emerged as the top three states by total transaction value.
- Investors aged 26-35 contributed the highest total SIP investment amount among all age groups.

This report documents the methodology, findings, and recommendations arising from the analysis, supported by the interactive Power BI dashboard built as part of this project.

2. Data Sources

The analysis was built on ten structured datasets covering different aspects of the mutual fund ecosystem. All data was provided in CSV format and represents a realistic simulation of the Indian mutual fund market.

Dataset	Description	Records
Fund Master	Scheme details: category, expense ratio, risk grade, fund manager	40
NAV History	Daily Net Asset Value for each scheme, Jan 2022 - May 2026	46,000
AUM by Fund House	Quarterly Assets Under Management per fund house	90
Monthly SIP Inflows	Industry-wide SIP contribution and account data	48
Category Inflows	Net inflow/outflow by fund category per month	144
Industry Folio Count	Total investor folio counts by fund type	21
Scheme Performance	Returns, alpha, beta, Sharpe, Sortino, drawdown metrics	40
Investor Transactions	Individual SIP, lumpsum and redemption transactions	32,778
Portfolio Holdings	Stock-level holdings and sector weights per scheme	322
Benchmark Indices	Daily closing values for Nifty 50, Nifty 100 and other indices	8,050

In addition to the static datasets, live NAV data was fetched directly from the public mfapi.in API for six benchmark schemes (HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip and Kotak Bluechip) to validate that the project pipeline could also ingest real-time data sources, not only static files.

3. Project Methodology

The project was structured into seven sequential phases over one week, each with clearly defined tasks and deliverables. This phased approach mirrors how a professional data analytics engagement would typically be planned and executed.

Phase	Focus Area	Key Outputs
Day 1	Project Setup and Data Ingestion	Project folder structure, Git repository, requirements.txt, data_ingestion.py, live_nav_fetch.py
Day 2	Data Cleaning and Database Design	Cleaned CSVs, SQLite star schema, 5 fact/dimension tables, 10 analytical SQL queries, data dictionary
Day 3	Exploratory Data Analysis	EDA_Analysis.ipynb with 9+ charts covering NAV trends, AUM, SIP, demographics and sectors
Day 4	Fund Performance Analytics	CAGR, Sharpe, Sortino, Alpha/Beta, Maximum Drawdown, Fund Scorecard, benchmark comparison
Day 5	Dashboard Development	4-page interactive Power BI dashboard with slicers, relationships and KPI cards
Day 6	Advanced Analytics and Risk	VaR/CVaR, rolling Sharpe ratio, cohort analysis, SIP continuity, fund recommender, sector HHI
Day 7	Final Report and Presentation	This report, project presentation, cleaned codebase, README, and tagged GitHub release

Throughout each phase, work was committed to a GitHub repository with descriptive commit messages, ensuring a transparent and traceable history of the project's development from raw data to final deliverables.

3.1 Tools and Technologies

- Python (pandas, numpy, scipy) for data cleaning, transformation and statistical analysis
- SQLite and SQLAlchemy for structured data storage and SQL-based querying
- Matplotlib, Seaborn and Plotly for static and interactive data visualisation
- Jupyter Notebook as the primary environment for exploratory and analytical work
- Power BI Desktop for the final interactive dashboard
- Git and GitHub for version control and project history

4. ETL Design and Data Architecture

The project followed a structured Extract-Transform-Load (ETL) pipeline to move data from raw CSV files into a query-ready analytical database.

3.1 Extract

All ten CSV files were loaded using Pandas. Each file's shape, column data types and first rows were printed and reviewed to understand structure and identify anomalies such as inconsistent date formats and missing values. A separate script connected to the mfapi.in REST API to fetch live NAV history in JSON format and convert it into CSV for consistency with the rest of the pipeline.

3.2 Transform

Data cleaning addressed several issues found during exploration:

- NAV History: dates parsed to proper datetime format, sorted by scheme and date, missing NAVs forward-filled to handle weekends and holidays, duplicate records removed, and NAV values validated to be greater than zero.
- Investor Transactions: transaction_type values standardised to uppercase (SIP, LUMPSUM, REDEMPTION), transaction amounts validated to be positive, and dates converted to a consistent format.
- Scheme Performance: all return columns converted to numeric type, with expense ratios checked against an expected business range of 0.1% to 2.5% to flag any anomalies.

3.3 Load

Cleaned datasets were loaded into a SQLite database (bluestock_mf.db) using SQLAlchemy's create_engine and to_sql functions. The database follows a star schema design with one dimension table (dim_fund) and four fact tables (fact_nav, fact_transactions, fact_performance, fact_aum), connected through the amfi_code key. This design allows efficient joins between fund attributes and time-series or transactional data.

Row counts after loading were verified against the source CSV files to confirm no records were lost during the pipeline: 40 schemes, 46,000 NAV records, 32,778 transactions, and 90 AUM records all matched their source counts exactly.

5. Exploratory Data Analysis Findings

Exploratory analysis was carried out across NAV trends, AUM growth, SIP flows, category inflows, investor demographics, geography, folio growth, return correlation, and portfolio sector allocation. The key findings are summarised below.

4.1 NAV Trends and Market Cycles

NAV trends for the top funds by AUM showed a clear bull run through 2023 followed by a market correction phase in 2024, consistent with broader market cycles. Despite the correction, most top-performing schemes recovered and continued an upward trend into 2025 and 2026.

4.2 AUM Growth and Fund House Dominance

SBI Mutual Fund consistently held the largest average AUM among the fund houses studied, followed by ICICI Prudential Mutual Fund and HDFC Mutual Fund. Total industry AUM (lakh crore) rose sharply between 2022 and 2024 before flattening slightly in 2025, broadly tracking the equity market cycle.

4.3 SIP Inflow Growth

Monthly SIP inflows grew steadily from ₹11,517 crore in January 2022 to an all-time high of ₹31,002 crore in December 2025. This consistent upward trend, even through the 2024 market correction, indicates strong retail investor discipline and growing confidence in systematic investing.

4.4 Category-wise Inflows

Equity-oriented categories attracted the strongest net inflows over the period studied, with Liquid funds also showing very high inflow volumes due to their use as short-term parking instruments by institutional and retail investors alike.

4.5 Investor Demographics

Investors in the 26-35 age group contributed the largest total transaction value, followed by the 36-45 group. Lumpsum investments accounted for the largest share of total transaction value (58.49%), followed by Redemptions, with SIP transactions being the most frequent in count but smaller in average ticket size.

4.6 Geographic Distribution

Punjab, Tamil Nadu and Madhya Pradesh were the top three states by total investment amount, with the leading ten states together accounting for a substantial majority of all transaction value in the dataset.

4.7 Folio Growth

Total investor folios grew from 13.26 crore in January 2022 to 26.12 crore in December 2025, an increase of nearly 97%, reflecting the continued mainstreaming of mutual fund investing in India.

4.8 Return Correlation and Sector Concentration

Daily return correlation among the top ten funds was generally high and positive, indicating that most equity-oriented schemes move together with broad market direction. Sector allocation analysis across portfolio holdings showed Financial Services and Information Technology as the dominant sectors across equity scheme portfolios.

6. Fund Performance Analysis

Performance analytics were computed for all 40 schemes using standard quantitative finance metrics, allowing funds to be compared on a like-for-like, risk-adjusted basis rather than on raw returns alone.

5.1 Returns and CAGR

Compound Annual Growth Rate (CAGR) was calculated for 1, 3 and 5 year horizons for every scheme. Small Cap and Mid Cap equity funds delivered the highest 1-year and 3-year CAGR figures, while Liquid and Debt-oriented schemes delivered lower but far more stable returns.

5.2 Risk-Adjusted Metrics

Sharpe Ratio (using a 6.5% risk-free rate as an RBI repo rate proxy) and Sortino Ratio were calculated for all schemes. Among growth-oriented equity funds, ABSL Small Cap Fund and SBI Small Cap Fund ranked highest on 1-year returns combined with strong Sharpe ratios near 0.90-0.94, indicating that their returns were earned with comparatively reasonable risk.

Liquid and short-duration debt funds showed unusually high Sharpe ratios in absolute terms because of their very low volatility, but this reflects their role as capital-preservation instruments rather than growth vehicles, and should be interpreted in that context rather than compared directly with equity fund Sharpe ratios.

5.3 Alpha and Beta

Alpha and Beta were estimated using linear regression of each fund's daily returns against the Nifty 100 benchmark. Funds with positive alpha demonstrated genuine value addition by the fund manager beyond what market exposure alone would explain, while Beta values close to 1.0 indicated funds that move broadly in line with the market, and values above 1.0 indicated higher sensitivity to market swings.

5.4 Maximum Drawdown

Maximum drawdown analysis identified the worst peak-to-trough decline experienced by each scheme. As expected, small and mid cap equity funds experienced deeper drawdowns than large cap or debt-oriented schemes, particularly during the 2024 correction period, reinforcing the standard risk-return trade-off.

5.5 Fund Scorecard

A composite Fund Scorecard (0-100) was built combining 3-year return rank (30%), Sharpe ratio rank (25%), Alpha rank (20%), expense ratio rank (15%, lower cost favoured) and maximum drawdown rank (10%, lower drawdown favoured). This scorecard allows funds to be ranked on a single, balanced measure rather than any one metric in isolation, and is provided in full in the accompanying fund_scorecard.csv file.

5.6 Benchmark Comparison

The top 5 scorecard-ranked funds were plotted against the Nifty 50 and Nifty 100 indices over a trailing 3-year window, normalised to a common starting base of 100. Most of the top-ranked funds outperformed both benchmarks over the period, with tracking error calculated as the annualised standard deviation of the difference between fund and benchmark daily returns.

7. Interactive Dashboard

A four-page interactive Power BI dashboard was built on top of the cleaned dataset, with relationships established between the fund master table and each fact table on the amfi_code key. The dashboard allows stakeholders to explore the data visually and interactively, using slicers to filter by fund house, category, plan, state, age group and city tier.

7.1 Page 1 - Industry Overview

This page presents headline KPIs (total AUM, SIP inflows, total folios) alongside an AUM growth trend line from 2022-2025 and a bar chart ranking fund houses by total AUM, clearly showing SBI Mutual Fund's market leadership.

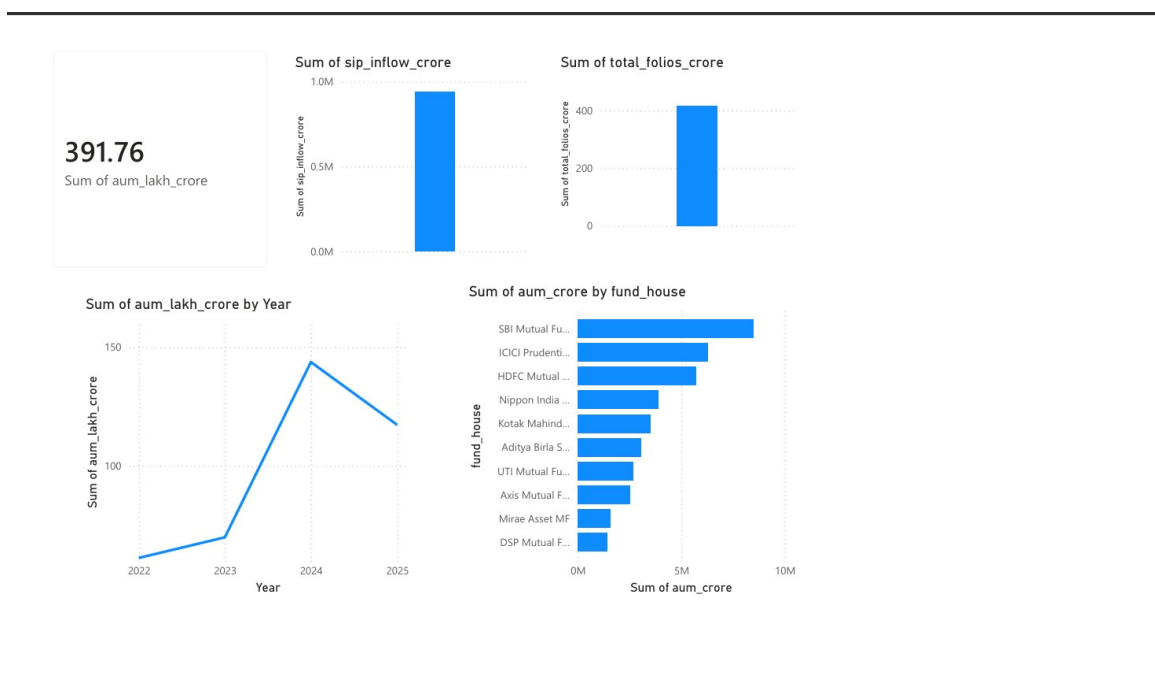


Figure 1: Industry Overview dashboard page - KPI cards, AUM trend, and AUM by fund house

7.2 Page 2 - Fund Performance

This page features a risk-return scatter plot (3-year return vs annualised standard deviation, bubble sized by AUM) alongside a full, sortable fund scorecard table with score, rank, CAGR and Sharpe ratio for every scheme, filterable by fund house, category and plan.

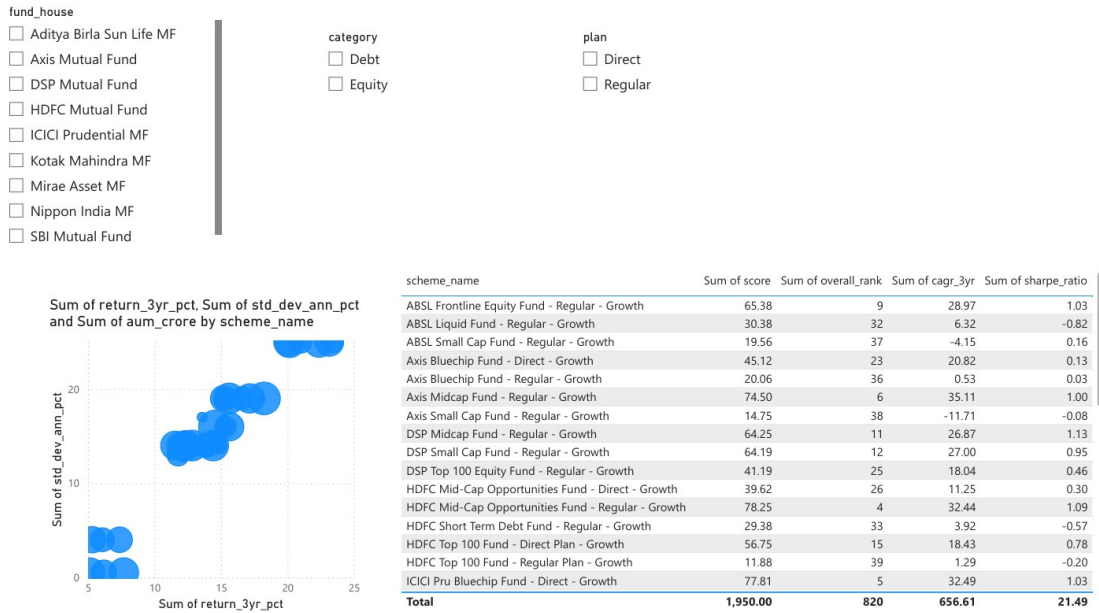


Figure 2: Fund Performance dashboard page - risk/return scatter plot and scorecard table

7.3 Page 3 - Investor Analytics

This page breaks down investor behaviour by transaction type (donut chart), by state (horizontal bar chart), and by age group, with slicers for state, age group and city tier enabling deep demographic drill-down.

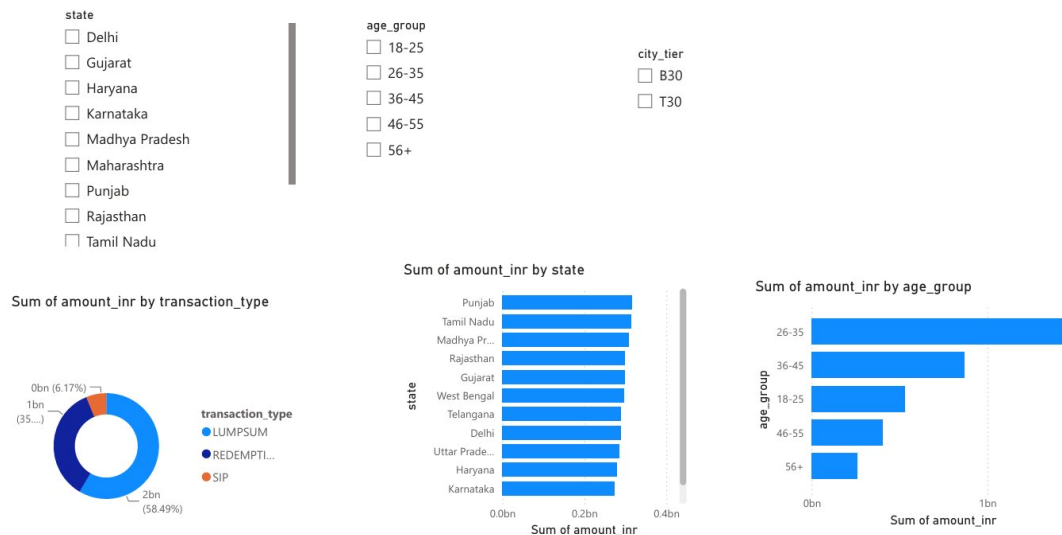


Figure 3: Investor Analytics dashboard page - transaction type split, state and age group breakdowns

7.4 Page 4 - SIP and Market Trends

This page tracks the monthly SIP inflow trend, net inflow by fund category, and total folio growth over time, giving a market-level view of how systematic investing and investor participation have evolved.

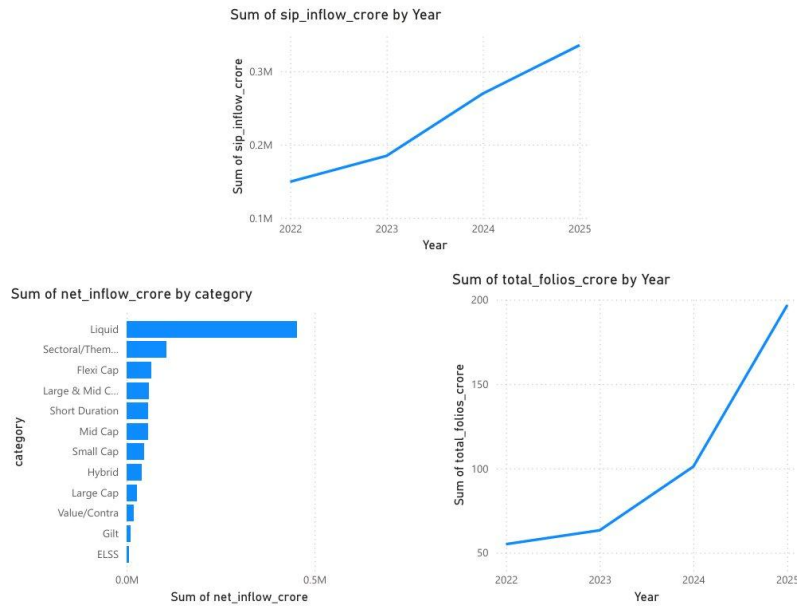


Figure 4: SIP and Market Trends dashboard page - SIP growth, category inflows, and folio count trend

8. Advanced Analytics and Risk Metrics

Beyond standard performance metrics, additional risk and behavioural analytics were carried out to provide a deeper view of fund risk and investor patterns.

8.1 Value at Risk (VaR) and Conditional VaR (CVaR)

Historical VaR at the 95% confidence level was computed as the 5th percentile of each fund's daily return distribution, with CVaR computed as the average return on days worse than the VaR threshold. This identifies, for every scheme, the magnitude of loss that can be expected on a typical bad day (VaR) versus an extreme bad day (CVaR), and is reported in full in `var_cvar_report.csv`.

8.2 Rolling 90-Day Sharpe Ratio

A rolling 90-day Sharpe ratio was calculated and plotted for the five largest funds by AUM, revealing how risk-adjusted performance shifted across the 2023 bull run and 2024 correction. Most funds showed Sharpe ratios comfortably above 1.0 during the bull run, dipping notably during the correction period before partially recovering.

8.3 Investor Cohort Analysis

Investors were grouped into cohorts based on the year of their first transaction. Each cohort's total invested amount, average SIP size, and most preferred fund were computed, showing how investor behaviour and fund preference have shifted as the investor base has grown year on year.

8.4 SIP Continuity Analysis

For investors with six or more SIP transactions, the average gap between consecutive SIP dates was computed. Investors with an average gap greater than 35 days were flagged as 'at-risk' of discontinuing their SIP, providing a practical early-warning indicator that could support retention campaigns in a real business setting.

8.5 Fund Recommender

A simple rules-based fund recommender was built (`recommender.py`) that takes a risk appetite input (Low, Moderate or High), maps it to the corresponding `risk_grade` values in the dataset, and returns the top 3 funds by Sharpe ratio within that risk band. For example, for a High risk appetite, the recommender returned Kotak Emerging Equity Fund, ICICI Pru Midcap Fund and SBI Small Cap Fund as the top three matches.

8.6 Sector Concentration (HHI)

The Herfindahl-Hirschman Index (HHI), calculated as the sum of squared sector weights within each fund's portfolio, was used to measure how concentrated or diversified each equity fund's sector exposure is. Funds with a small number of dominant sector bets showed materially higher HHI scores than broadly diversified funds, which is a useful input when assessing concentration risk independent of headline returns.

9. Limitations

- The dataset, while realistic in structure, is a simulated representation of the mutual fund industry and not live regulatory AMFI data; absolute figures should be read as illustrative rather than official industry statistics.
- Risk-free rate was assumed constant at 6.5% across the full period for Sharpe and Sortino calculations, whereas the actual RBI repo rate has varied over this time and would refine these metrics further.
- Sharpe ratio comparisons across very different fund categories (e.g., Liquid funds vs Small Cap equity funds) should be interpreted carefully, since low-volatility categories naturally produce higher Sharpe values that are not directly comparable to growth-oriented categories.
- The fund recommender uses a simple rules-based ranking by Sharpe ratio within a risk band; it does not account for an investor's investment horizon, tax situation, or existing portfolio composition.
- Cohort and continuity analysis used a 500-investor sample for the SIP continuity check to keep processing efficient; a full production run would process the entire investor base.
- The Power BI dashboard was built and exported using Power BI Desktop's free tier and was not published to the Power BI Service, so it is shared via static screenshots and the .pbix file rather than a live web URL.

10. Recommendations

- Encourage continued SIP adoption: given the strong and consistent SIP growth trend, fund houses and distributors should continue promoting SIPs, particularly to the 18-25 and 46-55 age segments that currently contribute less than the dominant 26-45 cohort.
- Target at-risk SIP investors proactively: the SIP continuity flagging logic developed in this project could be operationalised as a recurring monthly job to identify investors whose SIP gap has grown beyond 35 days, enabling timely retention outreach before a SIP lapses entirely.
- Use the Fund Scorecard for balanced fund selection: rather than ranking funds on trailing returns alone, the composite scorecard developed here (combining returns, Sharpe ratio, alpha, expense ratio and drawdown) offers a more balanced basis for fund recommendation and internal fund house comparison.
- Expand geographic outreach beyond top states: while Punjab, Tamil Nadu and Madhya Pradesh lead in investment value, the gap to other large states is relatively narrow, suggesting room for targeted growth campaigns in adjacent states with currently lower penetration.
- Monitor sector concentration risk: equity funds with high HHI scores carry concentration risk that may not be visible from returns alone; this should be factored into ongoing fund risk reviews alongside standard volatility metrics.
- Extend the Value at Risk analysis: incorporating a longer history and stress-testing scenarios (e.g., a simulated repeat of the 2024 correction) would further strengthen the risk monitoring framework built in this project.

11. Conclusion

This capstone project successfully delivered a complete, end-to-end mutual fund analytics pipeline: from raw data ingestion and cleaning, through a structured SQLite database, to exploratory analysis, quantitative performance metrics, an interactive Power BI dashboard, and advanced risk and behavioural analytics. The findings consistently point to a maturing Indian mutual fund market characterised by strong and resilient SIP growth, a steadily expanding investor base, and clear differentiation in risk-return profiles across fund categories.

The tools, scripts and dashboard produced as part of this project (data_ingestion.py, live_nav_fetch.py, the SQLite database and schema, the EDA, Performance Analytics and Advanced Analytics notebooks, the fund scorecard, and the four-page Power BI dashboard) together form a reusable analytical framework that could be extended with live data feeds, additional fund houses, or a production-grade recommendation engine in future work.

Appendix A: Glossary of Key Metrics

Term	Definition
CAGR	Compound Annual Growth Rate - the smoothed annual rate of growth of an investment between a start and end value over a given period.
Sharpe Ratio	Return earned in excess of the risk-free rate, divided by the standard deviation of returns; measures return per unit of total risk.
Sortino Ratio	Similar to the Sharpe Ratio but uses only downside (negative) return volatility in the denominator, focusing purely on harmful risk.
Alpha	The excess return of a fund relative to what would be expected given its Beta and the benchmark's return; positive alpha indicates manager outperformance.
Beta	A measure of a fund's sensitivity to benchmark movements; a Beta of 1.0 means the fund moves in line with the market.
Maximum Drawdown	The largest peak-to-trough decline in NAV observed over the analysis period, expressed as a percentage.
Value at Risk (VaR)	The loss level that is not expected to be exceeded with a given confidence level (95% in this report) over a one-day horizon.
Conditional VaR (CVaR)	The average loss in the worst-case scenarios beyond the VaR threshold; also known as Expected Shortfall.
HHI	Herfindahl-Hirschman Index - sum of squared portfolio weights, used here at the sector level to measure portfolio concentration.
AUM	Assets Under Management - the total market value of investments managed by a fund or fund house.
Folio	A unique account number assigned to an investor's holding in a mutual fund scheme, used as a proxy for investor count.
Tracking Error	The annualised standard deviation of the difference between a fund's returns and its benchmark's returns.

Appendix B: Project Repository Structure

The complete project codebase, notebooks, database, dashboard and reports are organised in the following structure and published to GitHub:

```

bluestock_mf_capstone/
  data/raw/ - original source CSV files
  data/processed/ - cleaned CSVs, fund scorecard, VaR report
  data/db/ - bluestock_mf.db (SQLite database)
  notebooks/ - EDA_Analysis.ipynb, Performance_Analytics.ipynb,
Advanced_Analytics.ipynb
  scripts/ - data_ingestion.py, live_nav_fetch.py, data_cleaning.py,
create_database.py, recommender.py
  sql/ - schema.sql, queries.sql

```

dashboard/ - bluestock_mf_dashboard.pbix, page screenshots
 reports/ - chart exports, Final_Report.pdf, Bluestock_MF_Presentation.pptx
 README.md - project overview and setup instructions

Appendix C: Self-Review Checklist

Checklist Item	Status
All 10 datasets ingested and validated	Complete
Live NAV API integration tested	Complete
SQLite database with star schema built	Complete
10+ analytical SQL queries written	Complete
EDA notebook with 9+ charts	Complete
Fund scorecard and performance metrics computed	Complete
4-page Power BI dashboard with slicers	Complete
VaR, CVaR, cohort and recommender analytics	Complete
Final report and presentation produced	Complete
GitHub repository with README and version tag	Complete